Computer Networking

Vlan

Class Material



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Version 1.0

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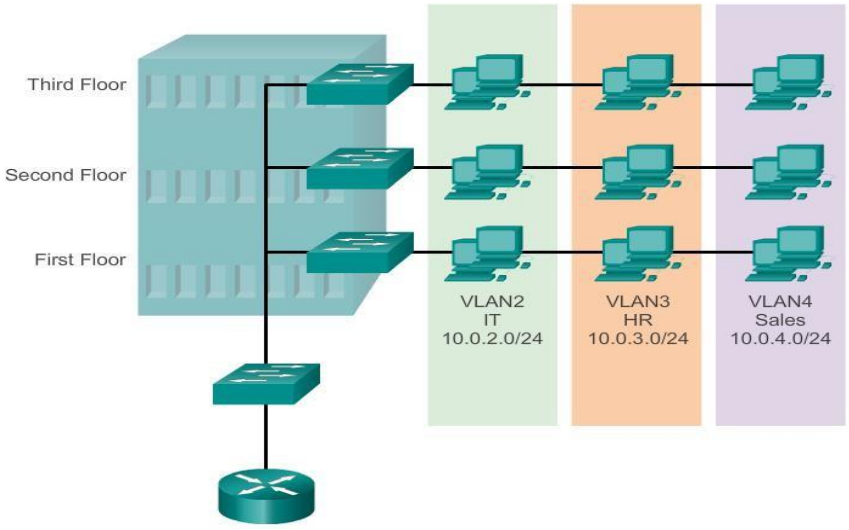
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Leaning Objectives

- Explain the purpose of VLAN in a switched network
- Analyze how a switch forwards frames based VLAN configuration in a multi-switched environment
- Configure a switch port to be assigned to a VLAN based on requirements
- Configure a trunk port on a LAN switch
- Troubleshoot VLAN and trunk configurations in a switched network

1| What is Addressing

A Virtual Local Area Network (VLAN) is a Layer 2 technique that logically segments a single physical network switch into multiple distinct broadcast domains. By grouping devices virtually, VLANs improve security, enhance performance, and simplify network management, allowing isolated networks (e.g., Guest, IoT, Office) to coexist on the same hardware



2| Type of Vlan

- **Data VLAN**
- **Default VLAN**
- **Native VLAN**
- **Management VLAN**

VLAN 1

```
Switch# show vlan brief

VLAN Name                 Status    Ports
---- -------------------- --------- -----------------------
1    default              active    Fa0/1,  Fa0/2,  Fa0/3,  Fa0/4
                                    Fa0/5,  Fa0/6,  Fa0/7,  Fa0/8
                                    Fa0/9,  Fa0/10, Fa0/11, Fa0/12
                                    Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                    Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                    Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                    Gi0/1,  Gi0/2
1002 fddi-default         act/unsup
1003 token-ring-default   act/unsup
1004 fddinet-default      act/unsup
1005 trnet-default        act/unsup
```

- All ports assigned to VLAN 1 to forward data by default.
- Native VLAN is VLAN 1 by default.
- Management VLAN is VLAN 1 by default.
- VLAN 1 cannot be renamed or deleted.

VLAN Assignment-VLAN Ranges On Catalyst Switches

- The Catalyst 2960 and 3560 Series switches support over 4,000 VLANs
- These VLANs are split into 2 categories:
- Normal Range VLANs
  - VLAN numbers from 1 through 1005
  - Configurations stored in the vlan.dat (in the flash)
  - VTP can only learn and store normal range VLANs
- Extended Range VLANs
  - VLAN numbers from 1006 through 4096
  - Configurations stored in the running-config (in the NVRAM)
  - VTP does not learn extended range VLANs

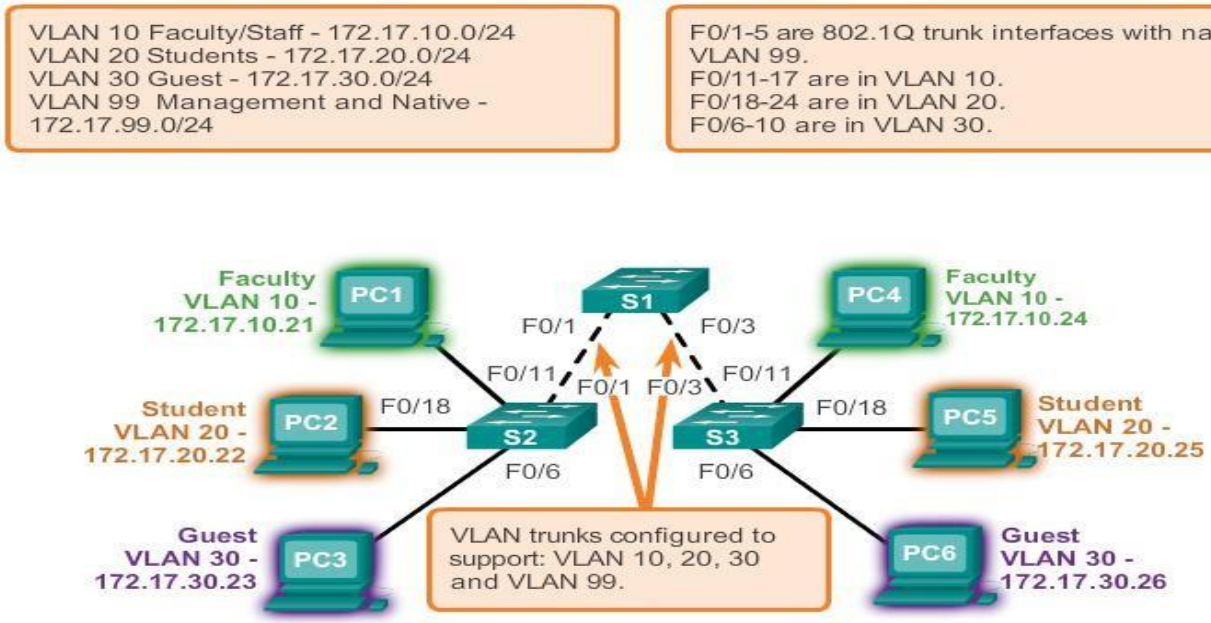
Creating a VLAN

Cisco Switch IOS Commands	
Enter global configuration mode.	`S1# configure terminal`
Create a VLAN with a valid id number.	`S1(config)# vlan vlan_id`
Specify a unique name to identify the VLAN.	`S1(config)# name vlan_name`
Return to the privileged EXEC mode.	`S1(config)# end`

Assigning Ports to VLAN

Cisco Switch IOS Commands	
Enter global configuration mode.	S1 # **configure terminal**
Enter interface configuration mode for the SVI.	S1(config) # **interface** *interface_id*
Configure the management interface IP address.	S1(config) # **ip address 172.17.99.11**
Set the port to access mode.	S1(config-if) # **switchport mode access**
Assign the port to a VLAN.	S1(config-if) # **switchport access vlan** *vlan_id*
Return to the privileged EXEC mode.	S1(config-if) # **end**

VLAN Trunks



- A VLAN trunk carries more than one VLAN
- Usually established between switches so same-VLAN devices can communicate even if physically connected to different switches
- A VLAN trunk is not associated to any VLANs. Neither is the trunk ports used to establish the trunk link
- Cisco IOS supports IEEE802.1q, a popular VLAN trunk protocol

Benefits of VLANs

- Security
- Cost reduction
- Better performance
- Shrink broadcast domains
- Improved IT staff efficiency
- Simpler project and application management

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